

DTU



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Defining the object of assessment

Learning objectives of this video

- Describe the central properties of a product or system
- Define the object of assessment - the functional unit

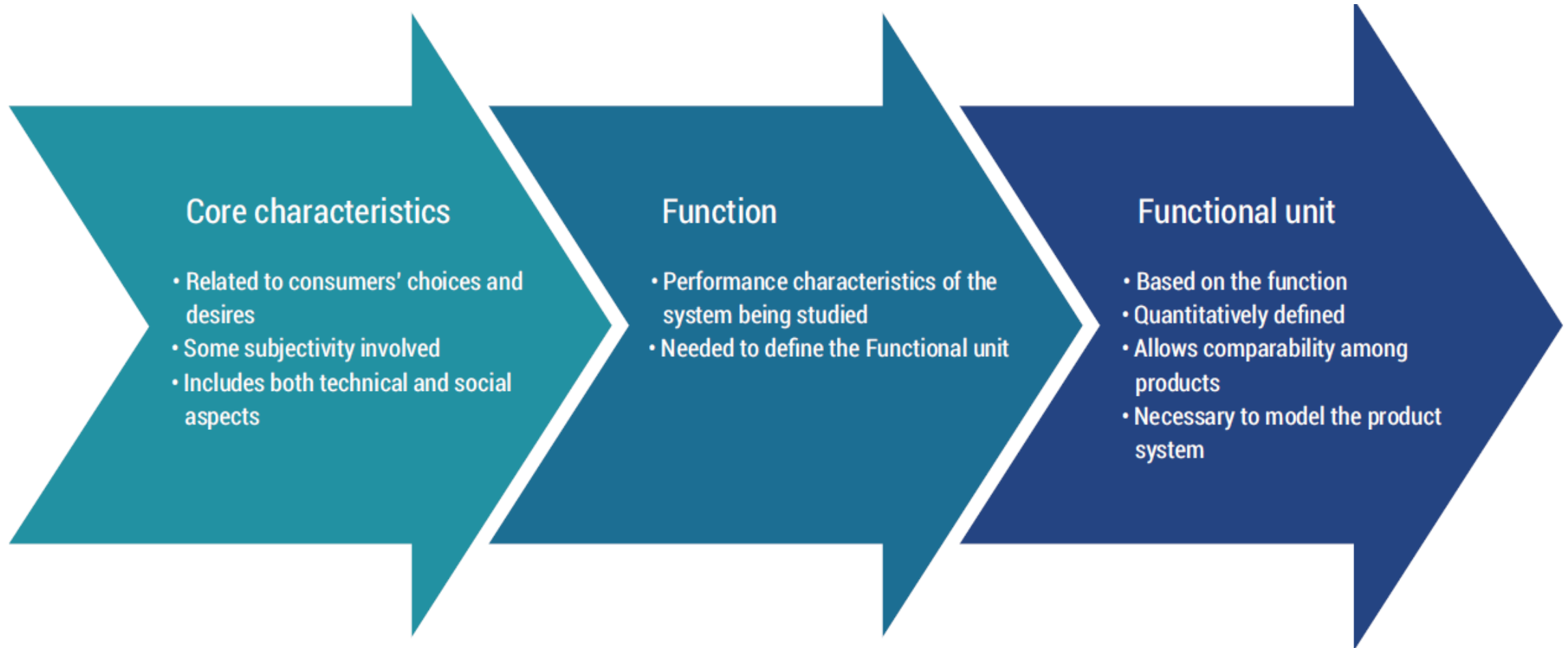
The use context sets the background for defining our product or system

- ▶ *“What should the product be used for?”*
- ▶ *“What does the product do?”*
- ▶ *“... For whom?”*
- ▶ *“... How long?” and “... how often?”*
- ▶ *“... Where in the world?”*
- ▶  *Which value contribution does the product deliver to the user.*

List the obligatory and positioning properties of the product or system you are assessing

- Obligatory properties:
 - Decisive for the customers perceiving the products as a product
 - Potential legal requirements
 - Helps define the product
- Positioning properties
 - Makes the product attractive to the customer and position the product in relation to comparable products

Defining the object of assessment is key to comparative studies



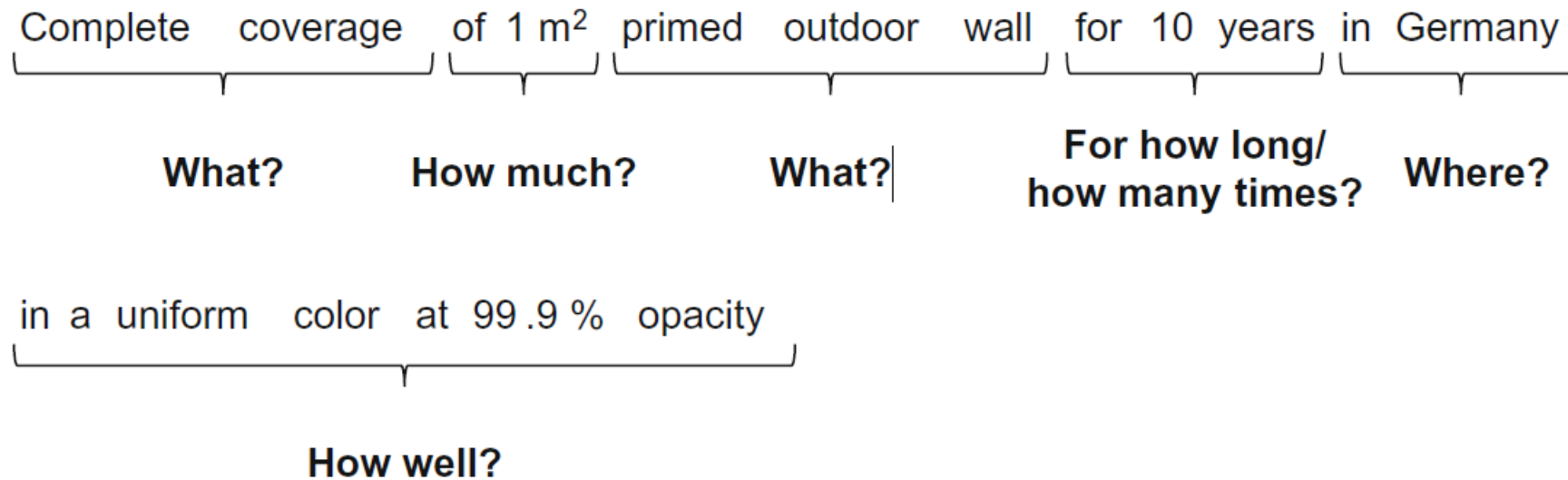
UNEP, 2020. *Guidelines for Social Life Cycle Assessment of Products and Organizations 2020*. Benoît Norris, C., et al (eds.). United Nations Environment Programme (UNEP) (15) (PDF) *Guidelines for Social Life Cycle Assessment of Products and Organizations 2020*.

Available from: <https://wedocs.unep.org/handle/20.500.11822/34554;jsessionid=550D7CE83D0AEDDC5362FC7FA33DA7AA>

Defining the functional unit – quantification of service/function/ use context - is essential

- The functional unit is the object for the analysis:
- Expresses the service the product/system deliver in quantitative and qualitative terms
 - Quantity (weight, volume, area, use frequency.....)
 - Duration of the service
- Central qualities or properties
- Essential in order to ensure equivalence between the compared services/systems

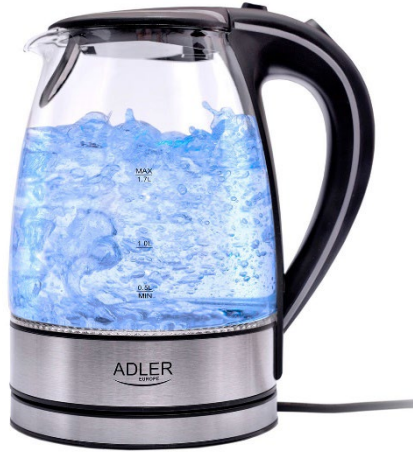
Example of definition of functional unit



*MZ Hauschild, RK Rosenbaum, SI Olsen: Life Cycle Assessment, Theory and Practice
DOI 10.1007/978-3-319-56475-3*

Exercise: Describe the functional unit

- Of a water boiler



- Of a lawn mower



Examples of functional units

- Water boiler:
 - Heat 1 liter of water to the boiling point three times a day for a year in Denmark
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